

MODEL QUESTION PAPER 1

S5 INSTRUMENTATION ENGINEERING**COURSE: ROBOTICS AND CONTROL***Time: 3 Hour**Max. Marks: 75****I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)***

		Module Outcomes	Cognitive Level
1	What is a robot?	M1.01	R
2	Name any two commonly used robot configuration system?	M1.03	U
3	What is mean by work volume?	M1.01	U
4	State any two advantages of a hydraulic drive.	M2.01	U
5	What is meant by step angle of a stepper motor?	M2.04	U
6	List any two common imaging device used for robot vision system?	M3.01	U
7	What is meant by a tactile sensor?	M3.03	U
8	What is mean by automation?	M4.01	U
9	What are the gantry robots?	M4.03	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	Write Asimov's laws of robotics?	M1.01	R
2.	What is meant by accuracy and precision of a robot?	M1.02	U
3.	What is meant by pitch, yaw and roll?	M1.02	U
4.	What are the properties of stepper motor?	M2.02	U
5.	List the types of encoders used in robotics drive system.	M2.04	U
6.	Explain the working of DC servomotor.	M2.02	U
7.	List the desirable properties of hydraulic fluid.	M2.01	U
8.	Distinguish between proximity and a range sensor?	M3.01	U
9.	List the applications of robot vision system.	M3.03	U
10.	What are the basic elements of any automated system?	M4.01	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	Explain the operation of cartesian and cylindrical configuration with neat sketches.	M1.03	U
IV	OR List the advantages and disadvantages of robots.	M1.04	U
V	What are the advantages and disadvantages of hydraulic drives?	M2.01	U
VI	OR Explain pneumatic drive system with a neat sketch.	M2.01	U
VII	With necessary figures explain the working of a stepper motor.	M2.02	U
VIII	OR List the advantages of using DC servo motor and stepper motor.	M2.02	U
IX	Explain the operations of a. a piezo electric sensor and b. a force sensor.	M3.01	U
X	OR Distinguish with example, the difference between proximity and range sensors	M3.01	U
XI	With necessary figure explain vidicon camera.	M3.03	U
XII	OR With necessary figure explain the operation of CCD.	M3.03	U
XIII	Explain the advanced automation functions.	M4.02	U
XIV	OR Explain the biomedical applications of robotics	M4.03	U

MODEL QUESTION PAPER 2**S5 INSTRUMENTATION ENGINEERING****COURSE: ROBOTICS AND AUTOMATION***Time: 3 Hour**Max. Marks: 75***I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)**

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	Define robot anatomy.	M1.02	U
2	What is mean by Power drives?	M2.01	U
3	List any two sensors in robotics.	M3.02	U
4	Define Automation.	M4.01	U
5	Classify the types of end effector.	M1.02	U
6	Define Tachometer.	M2.04	U
7	Mention any two applications of robot vision system.	M3.04	U
8	List any two applications of robots.	M4.03	U
9	What is mean by work volume?	M1.03	U

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Explain the degrees of freedom associated with robot wrist.	M1.01	U
2.	Compare electric and hydraulic drives in robotics.	M2.02	U
3.	Explain the working of an encoder.	M3.02	U
4.	What are the basic elements of any automated system.	M4.01	U
5.	Describe the characteristic of first- and second-generation robot.	M1.03	U
6.	Describe the operation of electric power drives.	M2.02	U
7.	What are the main functions of sensors in robots.	M3.01	U
8.	Explain the levels of automation.	M4.02	U
9.	Explain Asimov's law of robotics.	M1.01	U
10.	With neat diagram, explain the construction and working of optical	M2.04	U

	encoder.		
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Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	Explain the types of manipulator joints.	M1.02	U
	OR		
IV	Sketch and Explain the operation of cylindrical coordinate robot.	M1.03	U
V	What are the advantages and disadvantages of pneumatic drives.	M2.02	U
	OR		
VI	With necessary figures explain the working of a stepper motor.	M2.03	U
VII	Explain three functions of Machine vision system.	M3.03	U
	OR		
VIII	Explain the operations of range sensor.	M3.02	U
IX	Describe the reasons for automation.	M4.01	U
	OR		
X	Explain the biomedical applications of robotics.	M4.03	U
XI	Explain the construction and working of tactile sensor.	M3.02	U
	OR		
XII	Describe the working principle of proximity sensor.	M3.02	U
XIII	With necessary figures explain the working of DC tachometer.	M2.04	U
	OR		
XIV	List the advantages of using DC servo motor and stepper motor.	M2.03	U